

Derivation (3/3)

- Therefore, we get the discrete-time Fourier transform pair

Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Observations
 - ▶ Continuous spectrum: Similar to CTFT
 - ▶ Periodic with period 2π : Different from CTFT
 - ▶ Low frequency: close to 0 and 2π ; high frequency: close to π

为什么pi是高频?

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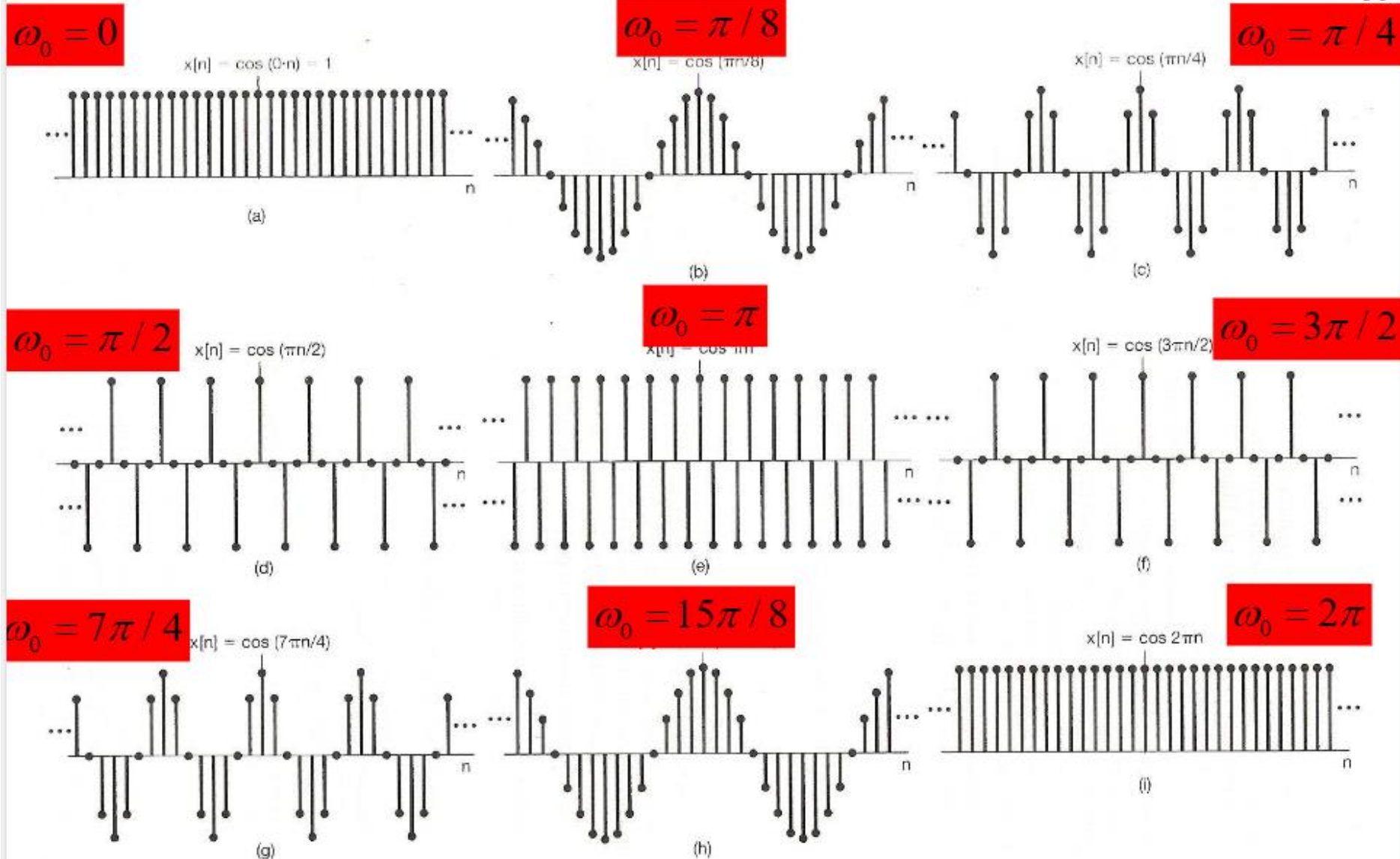


Figure 1.27 Discrete-time sinusoidal sequences for several different frequencies.

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$	$X(e^{j\omega})$ periodic with
		$y[n]$	$Y(e^{j\omega})$ period 2π
5.3.2	Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
5.3.5	Differencing in Time	$x[n] - x[n - 1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - e^{-j\omega}} X(e^{j\omega})$
			$+ \pi X(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \Re\{X(e^{j\omega})\} = \Re\{X(e^{-j\omega})\} \\ \Im\{X(e^{j\omega})\} = -\Im\{X(e^{-j\omega})\} \\ X(e^{j\omega}) = X(e^{-j\omega}) \\ \angle X(e^{j\omega}) = -\angle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real and even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}\{x[n]\} \quad [x[n] \text{ real}]$ $x_o[n] = \mathcal{O}\{x[n]\} \quad [x[n] \text{ real}]$	$\Re\{X(e^{j\omega})\}$ $j\Im\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals		$\sum_{n=-\infty}^{+\infty} x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

TABLE 5.2 BASIC DISCRETE-TIME FOURIER TRANSFORM PAIRS

Signal	Fourier Transform	Fourier Series Coefficients (if periodic)
$\sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	a_k
$e^{j\omega_0 n}$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi l)$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} 1, & k = m, m \pm N, m \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\cos \omega_0 n$	$\pi \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} \frac{1}{2}, & k = \pm m, \pm m \pm N, \pm m \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi r}{N}$ $a_k = \begin{cases} \frac{1}{2j}, & k = r, r \pm N, r \pm 2N, \dots \\ -\frac{1}{2j}, & k = -r, -r \pm N, -r \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$x[n] = 1$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - 2\pi l)$	$a_k = \begin{cases} 1, & k = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$
Periodic square wave $x[n] = \begin{cases} 1, & n \leq N_1 \\ 0, & N_1 < n \leq N/2 \end{cases}$ and $x[n + N] = x[n]$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{\sin[(2\pi k/N)(N_1 + \frac{1}{2})]}{N \sin[2\pi k/2N]}, \quad k \neq 0, \pm N, \pm 2N, \dots$ $a_k = \frac{2N_1 + 1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$
$\sum_{k=-\infty}^{+\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{1}{N}$ for all k

$a^n u[n], \quad a < 1$	$\frac{1}{1 - ae^{-j\omega}}$	—
$x[n] \begin{cases} 1, & n \leq N_1 \\ 0, & n > N_1 \end{cases}$	$\frac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$	—
$\frac{\sin Wn}{\pi n} = \frac{W}{\pi} \text{sinc}\left(\frac{Wn}{\pi}\right)$ $0 < W < \pi$	$X(\omega) = \begin{cases} 1, & 0 \leq \omega \leq W \\ 0, & W < \omega \leq \pi \end{cases}$ $X(\omega)$ periodic with period 2π	—
$\delta[n]$	1	—
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{+\infty} \pi \delta(\omega - 2\pi k)$	—
$\delta[n - n_0]$	$e^{-j\omega n_0}$	—
$(n + 1)a^n u[n], \quad a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$	—
$\frac{(n + r - 1)!}{n!(r - 1)!} a^n u[n], \quad a < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$	—

5.1. Use the Fourier transform analysis equation (5.9) to calculate the Fourier transforms of:

(a) $(\frac{1}{2})^{n-1}u[n-1]$ **(b)** $(\frac{1}{2})^{|n-1|}$

Sketch and label one period of the magnitude of each Fourier transform.

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

5.1. Use the Fourier transform analysis equation (5.9) to calculate the Fourier transforms of:

(a) $(\frac{1}{2})^{n-1}u[n-1]$ **(b)** $(\frac{1}{2})^{|n-1|}$

Sketch and label one period of the magnitude of each Fourier transform.

(a)

$$\begin{aligned} X(e^{jw}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-jwn} \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} e^{-jwn} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-jw(n+1)} \\ &= e^{-jw} \frac{1}{(1 - (1/2)e^{-jw})} \end{aligned}$$

(b)

$$\begin{aligned} X(e^{jw}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-jwn} = \sum_{n=-\infty}^0 \left(\frac{1}{2}\right)^{-(n-1)} e^{-jwn} + \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} e^{-jwn} \\ \sum_{n=-\infty}^0 \left(\frac{1}{2}\right)^{-(n-1)} e^{-jwn} &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{n+1} e^{jwn} = \frac{1}{2} \frac{1}{(1 - (1/2)e^{jw})} \\ X(e^{jw}) &= \frac{1}{2} \frac{1}{(1 - (1/2)e^{jw})} + e^{-jw} \frac{1}{(1 - (1/2)e^{-jw})} = \frac{0.75e^{-jw}}{(1.25 - \cos w)} \end{aligned}$$

5.3. Determine the Fourier transform for $-\pi \leq \omega < \pi$ in the case of each of the following periodic signals:

(a) $\sin(\frac{\pi}{3}n + \frac{\pi}{4})$ **(b)** $2 + \cos(\frac{\pi}{6}n + \frac{\pi}{8})$

Fourier Transform of $e^{j\omega n}$

The following transform pair is actually NOT rigorously defined:

$$x[n] = e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)$$

- According to the Fourier series, for a periodic signal with period N :

$$x[n] = a_0 + a_1 e^{j(2\pi/N)n} + \dots + a_k e^{jk(2\pi/N)n} + \dots + a_{N-1} e^{j(N-1)(2\pi/N)n}$$

- $e^{jk(2\pi/N)n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - k(2\pi/N) - 2\pi l)$

5.3. Determine the Fourier transform for $-\pi \leq \omega < \pi$ in the case of each of the following periodic signals:

(a) $\sin(\frac{\pi}{3}n + \frac{\pi}{4})$ **(b)** $2 + \cos(\frac{\pi}{6}n + \frac{\pi}{8})$

(a)

$$x[n] = \frac{1}{2j} e^{j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)} - \frac{1}{2j} e^{-j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)} = \frac{1}{2j} e^{j\frac{\pi}{4}} e^{j\frac{\pi}{3}n} - \frac{1}{2j} e^{-j\frac{\pi}{4}} e^{-j\frac{\pi}{3}n}$$

$$a_1 = \frac{1}{2j} e^{j\frac{\pi}{4}}, \quad a_{-1} = -\frac{1}{2j} e^{-j\frac{\pi}{4}}$$

$$X(e^{jw}) = 2\pi a_1 \delta\left(w - \frac{\pi}{3}\right) + 2\pi a_{-1} \delta\left(w + \frac{\pi}{3}\right)$$

$$= (\pi/j) \left\{ e^{j\frac{\pi}{4}} \delta\left(w - \frac{\pi}{3}\right) - e^{-j\frac{\pi}{4}} \delta\left(w + \frac{\pi}{3}\right) \right\}$$

5.3. Determine the Fourier transform for $-\pi \leq \omega < \pi$ in the case of each of the following periodic signals:

(a) $\sin(\frac{\pi}{3}n + \frac{\pi}{4})$ **(b)** $2 + \cos(\frac{\pi}{6}n + \frac{\pi}{8})$

(b)

$$x[n] = 2 + \frac{1}{2}e^{j\left(\frac{\pi}{6}n + \frac{\pi}{8}\right)} + \frac{1}{2}e^{-j\left(\frac{\pi}{6}n + \frac{\pi}{8}\right)} = 2 + \frac{1}{2}e^{j\frac{\pi}{8}}e^{j\frac{\pi}{6}n} + \frac{1}{2}e^{-j\frac{\pi}{8}}e^{-j\frac{\pi}{6}n}$$

$$a_0 = 2, \quad a_1 = \frac{1}{2}e^{j\frac{\pi}{8}}, \quad a_{-1} = \frac{1}{2}e^{-j\frac{\pi}{8}}$$

$$X(e^{jw}) = 2\pi a_0 \delta(w) + 2\pi a_1 \delta\left(w - \frac{\pi}{6}\right) + 2\pi a_{-1} \delta\left(w + \frac{\pi}{6}\right)$$

$$= 4\pi \delta(w) + \pi \left\{ e^{j\pi/8} \delta\left(w - \frac{\pi}{6}\right) + e^{-j\pi/8} \delta\left(w + \frac{\pi}{6}\right) \right\}$$

5.4. Use the Fourier transform synthesis equation (5.8) to determine the inverse Fourier transforms of:

(a) $X_1(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \{2\pi\delta(\omega - 2\pi k) + \pi\delta(\omega - \frac{\pi}{2} - 2\pi k) + \pi\delta(\omega + \frac{\pi}{2} - 2\pi k)\}$

(b) $X_2(e^{j\omega}) = \begin{cases} 2j, & 0 < \omega \leq \pi \\ -2j, & -\pi < \omega \leq 0 \end{cases}$

5.4. Use the Fourier transform synthesis equation (5.8) to determine the inverse Fourier transforms of:

(a) $X_1(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \{2\pi\delta(\omega - 2\pi k) + \pi\delta(\omega - \frac{\pi}{2} - 2\pi k) + \pi\delta(\omega + \frac{\pi}{2} - 2\pi k)\}$

(b) $X_2(e^{j\omega}) = \begin{cases} 2j, & 0 < \omega \leq \pi \\ -2j, & -\pi < \omega \leq 0 \end{cases}$

(a)

$$x_1[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{jw}) e^{jwn} dw$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} [2\pi\delta(w) + \pi\delta(w - \frac{\pi}{2}) + \pi\delta(w + \frac{\pi}{2})] e^{jwn} dw$$

$$= e^{j0} + \frac{1}{2} e^{j\frac{\pi}{2}n} + \frac{1}{2} e^{-j\frac{\pi}{2}n}$$

$$= 1 + \cos\left(\frac{\pi n}{2}\right)$$

5.4. Use the Fourier transform synthesis equation (5.8) to determine the inverse Fourier transforms of:

(a) $X_1(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \{2\pi\delta(\omega - 2\pi k) + \pi\delta(\omega - \frac{\pi}{2} - 2\pi k) + \pi\delta(\omega + \frac{\pi}{2} - 2\pi k)\}$

(b) $X_2(e^{j\omega}) = \begin{cases} 2j, & 0 < \omega \leq \pi \\ -2j, & -\pi < \omega \leq 0 \end{cases}$

(b)

$$\begin{aligned} x_2[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_2(e^{j\omega}) e^{j\omega n} d\omega \\ &= -\frac{1}{2\pi} \int_{-\pi}^0 2je^{j\omega n} d\omega + \frac{1}{2\pi} \int_0^{\pi} 2je^{j\omega n} d\omega \\ &= \frac{j}{\pi} \left[-\frac{1 - e^{-j\pi n}}{jn} + \frac{e^{j\pi n} - 1}{jn} \right] \\ &= \frac{2\cos(n\pi) - 2}{n\pi} = -\frac{4}{n\pi} \sin^2\left(\frac{n\pi}{2}\right) \end{aligned}$$

5.41 Let $\tilde{x}[n]$ be a periodic signal with period N . A finite-duration signal $x[n]$ is related to $\tilde{x}[n]$ through

$$x[n] = \begin{cases} \tilde{x}[n], & n_0 \leq n \leq n_0 + N - 1 \\ 0, & \text{otherwise} \end{cases},$$

for some integer n_0 . That is, $x[n]$ is equal to $\tilde{x}[n]$ over one period and zero elsewhere.

(a) If $\tilde{x}[n]$ has Fourier series coefficients a_k and $x[n]$ has Fourier transform $X(e^{j\omega})$, show that

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

regardless of the value of n_0 .

(b) Consider the following two signals:

$$x[n] = u[n] - u[n - 5]$$

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n - kN]$$

where N is a positive integer. Let a_k denote the Fourier coefficients of $\tilde{x}[n]$ and let $X(e^{j\omega})$ denote the Fourier transform of $x[n]$.

(i) Determine a closed-form expression for $X(e^{j\omega})$.

(ii) Using the result of part (i), determine an expression for the Fourier coefficients a_k .

Derivation (2/3)

- Look at the Fourier series of $\tilde{x}[n]$:

$$Na_k = \sum_{n=-N/2}^{N/2+1} \tilde{x}[n] e^{-jk(2\pi/N)n} = \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

- Define $X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\omega n}$, so that $Na_k = X(e^{jk\omega_0})$ ($\omega_0 = 2\pi/N$)
- Look at the synthesis equation:

$$\begin{aligned} \tilde{x}[n] &= \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} = \frac{1}{N} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0 \end{aligned}$$

- When $N \rightarrow \infty$, $\omega_0 \rightarrow d\omega$, $\sum \rightarrow \int$, $k\omega_0 \rightarrow \omega$, $\tilde{x}[n] \rightarrow x[n]$, thus,

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

5.41 Let $\tilde{x}[n]$ be a periodic signal with period N . A finite-duration signal $x[n]$ is related to $\tilde{x}[n]$ through

$$x[n] = \begin{cases} \tilde{x}[n], & n_0 \leq n \leq n_0 + N - 1 \\ 0, & \text{otherwise} \end{cases},$$

for some integer n_0 . That is, $x[n]$ is equal to $\tilde{x}[n]$ over one period and zero elsewhere.

(a) If $\tilde{x}[n]$ has Fourier series coefficients a_k and $x[n]$ has Fourier transform $X(e^{j\omega})$, show that

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

(a)

regardless of the value of n_0 .

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=n_0}^{n_0+N-1} x[n] e^{-j\omega n}$$

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{n=n_0}^{n_0+N-1} x[n] e^{-j\frac{2\pi k}{N}n}$$

$$a_k = \frac{1}{N} \sum_{\langle N \rangle} x[n] e^{-j\frac{2\pi k}{N}n} = \frac{1}{N} \sum_{n=n_0}^{n_0+N-1} x[n] e^{-j\frac{2\pi k}{N}n}$$

$$a_k = \frac{1}{N} X\left(e^{j\frac{2\pi k}{N}}\right)$$

(b) Consider the following two signals:

$$x[n] = u[n] - u[n - 5]$$

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n - kN]$$

where N is a positive integer. Let a_k denote the Fourier coefficients of $\tilde{x}[n]$ and let $X(e^{j\omega})$ denote the Fourier transform of $x[n]$.

- (i)** Determine a closed-form expression for $X(e^{j\omega})$.
- (ii)** Using the result of part (i), determine an expression for the Fourier coefficients a_k .

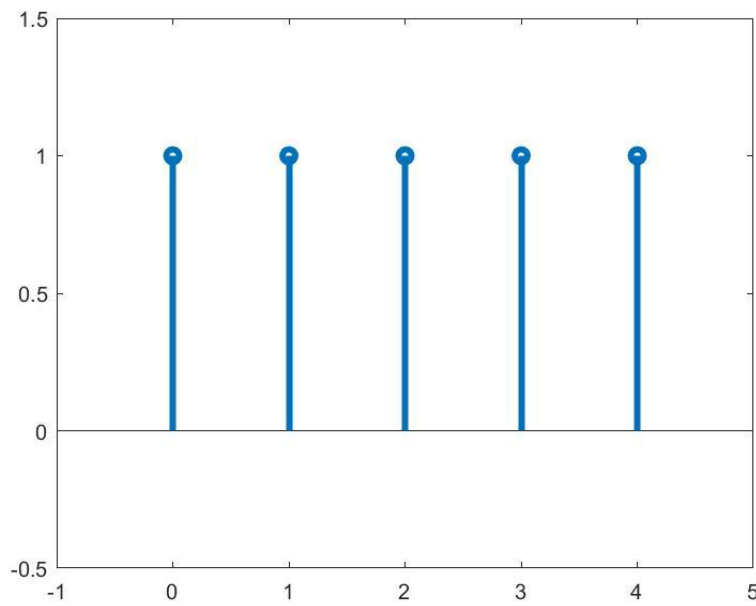
$$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(j\omega)$$

(i)

$$\begin{aligned} X(e^{j\omega}) &= 1 + e^{-j\omega} + e^{-j2\omega} + e^{-j3\omega} + e^{-j4\omega} \\ &= e^{-j2\omega} (e^{-j2\omega} + e^{-j\omega} + 1 + e^{j\omega} + e^{j2\omega}) \\ &= e^{-j2\omega} (2\cos(2\omega) + 2\cos(\omega) + 1) \end{aligned}$$

(ii)

$$a_k = \frac{1}{N} X\left(e^{j\frac{2\pi k}{N}}\right) = \frac{1}{5} e^{j\frac{4\pi k}{5}} \left(2\cos\left(\frac{4\pi k}{5}\right) + 2\cos\left(\frac{2\pi k}{5}\right) + 1\right)$$



CTFS

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_o t} dt \quad (\text{Analysis equation})$$

DTFS

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} \quad (\text{Analysis equation})$$

CTFT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Synthesis equation}$$

— "sum" of $e^{j\omega t}$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \quad \text{Analysis equation}$$

DTFT

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

时域

Continuous, periodic

Discrete, periodic

Continuous, aperiodic

Discrete, aperiodic

频域

Discrete, aperiodic

Discrete, periodic

Continuous, aperiodic

Continuous, periodic

Problem 5.1

5.1. Use the Fourier transform analysis equation (5.9) to calculate the Fourier transforms of:

(a) $(\frac{1}{2})^{n-1}u[n-1]$ (b) $(\frac{1}{2})^{|n-1|}$

Sketch and label one period of the magnitude of each Fourier transform.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$
$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

Answer 5.1



5.1 (a) let $x[n] = (1/2)^{n-1} u[n-1]$. Using the Fourier transform analysis equation (5.9), the Fourier transform $X(e^{j\omega})$ of this signal is

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ &= \sum_{n=1}^{\infty} (1/2)^{n-1} e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (1/2)^n e^{-j\omega(n+1)} \\ &= e^{-j\omega} \frac{1}{(1 - (1/2)e^{-j\omega})} \end{aligned}$$

Answer 5.1

(b) Let $x[n] = (1/2)^{|n-1|}$. Using the Fourier transform analysis equation (5.9). The Fourier transform $x(e^{jw})$ of signal is

$$x(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn} = \sum_{n=-\infty}^0 (1/2)^{-(n-1)} e^{-jwn} + \sum_{n=1}^{\infty} (1/2)^{n-1} e^{-jwn}$$

The second summation in the right—hand side of the above equation is exactly the same as result of part (a). Now ,

$$\sum_{n=-\infty}^0 (1/2)^{-(n-1)} e^{-jwn} = \sum_{n=0}^{\infty} (1/2)^{n+1} e^{jwn} = (1/2) \frac{1}{(1 - (1/2)e^{jw})}$$

Therefore

$$x(e^{jw}) = (1/2) \frac{1}{(1 - (1/2)e^{jw})} + e^{-jw} \frac{1}{(1 - (1/2)e^{-jw})} = \frac{0.75e^{-jw}}{(1.25 - \cos w)}$$

Problem 5.3

5.3. Determine the Fourier transform for $-\pi \leq \omega < \pi$ in the case of each of the following periodic signals:

(a) $\sin(\frac{\pi}{3}n + \frac{\pi}{4})$ (b) $2 + \cos(\frac{\pi}{6}n + \frac{\pi}{8})$

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - \frac{2\pi k}{N})$$

Answer 5.3

5.3 We note from section 5.2 that a periodic signal with Fourier series representation

$$x[n] = \sum_{k \in \langle N \rangle} a_k e^{jk(2\pi/N)n}$$

has a Fourier transform

$$X(e^{jw}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(w - \frac{2\pi k}{N})$$

(a) Consider the signal $x_1[n] = \sin(\frac{\pi}{3}n + \frac{\pi}{4})$. We note that the fundamental period of the signal $x_1[n]$ is $N=6$.

The signal may be written as

Answer 5.3

$$x_1[n] = \frac{1}{2j} e^{j(\frac{\pi}{3}n + \frac{\pi}{4})} - \frac{1}{2j} e^{-j(\frac{\pi}{3}n + \frac{\pi}{4})} = \frac{1}{2j} e^{j\frac{\pi}{4}} e^{j\frac{2\pi}{6}n} - \frac{1}{2j} e^{-j\frac{\pi}{4}} e^{-j\frac{2\pi}{6}n}$$

Form this, we obtain the non-zero Fourier series coefficients a_k of $x_1[n]$ the range $-2 \leq \pi \leq 3$ as

$$a_1 = (1/2j)e^{j\frac{\pi}{4}} \quad a_{-1} = -(1/2j)e^{-j\frac{\pi}{4}}$$

Therefore, in the range $-\pi \leq \omega \leq \pi$, we obtain

$$\begin{aligned} X(e^{j\omega}) &= 2\pi a_1 \delta(\omega - \frac{2\pi}{6}) + 2\pi a_{-1} \delta(\omega + \frac{2\pi}{6}) \\ &= (\pi/j) \{ e^{j\pi/4} \delta(\omega - 2\pi/6) - e^{-j\pi/4} \delta(\omega + 2\pi/6) \} \end{aligned}$$

Answer 5.3

(b) consider the signal $x_2[n] = 2 + \cos(\frac{\pi}{6}n + \frac{\pi}{8})$. we note that the fundamental period of the signal $x_1[n]$ is

$N=12$. the signal maybe written as

$$\begin{aligned} x_1[n] &= 2 + (1/2)e^{j(\frac{\pi}{6}n + \frac{\pi}{8})} + (1/2)e^{-j(\frac{\pi}{6}n + \frac{\pi}{8})} \\ &= 2 + (1/2)e^{j\frac{2\pi}{12}n} e^{j\frac{\pi}{8}} + (1/2)e^{-j\frac{2\pi}{12}n} e^{-j\frac{\pi}{8}} \end{aligned}$$

Form this ,we obtain the non-zero Fourier series coefficients a_k of $x_2[n]$ in the range $-5 \leq k \leq 6$ as

$$a_0 = 2 \quad a_1 = (1/2)e^{j\frac{\pi}{8}} \quad a_{-1} = (1/2)e^{-j\frac{\pi}{8}}$$

Therefore ,in the range ,we obtain

$$\begin{aligned} X(e^{jw}) &= 2\pi a_0 \delta(w) + 2\pi a_1 \delta(w - \frac{2\pi}{12}) + 2\pi a_{-1} \delta(w + \frac{2\pi}{12}) \\ &= 4\pi \delta(w) + \pi \{ e^{j\pi/8} \delta(w - \frac{\pi}{6}) + e^{-j\pi/8} \delta(w + \frac{\pi}{6}) \} \end{aligned}$$

Problem 5.4

5.4. Use the Fourier transform synthesis equation (5.8) to determine the inverse Fourier transforms of:

(a) $X_1(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \{2\pi\delta(\omega - 2\pi k) + \pi\delta(\omega - \frac{\pi}{2} - 2\pi k) + \pi\delta(\omega + \frac{\pi}{2} - 2\pi k)\}$

(b) $X_2(e^{j\omega}) = \begin{cases} 2j, & 0 < \omega \leq \pi \\ -2j, & -\pi < \omega \leq 0 \end{cases}$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Answer 5.4

5.4 (a) Using the Fourier transform synthesis equation (5.8)

$$\begin{aligned} x_1[n] &= (1/2\pi) \int_{-\pi}^{\pi} X_1(e^{jw}) e^{jwn} dw \\ &= (1/2\pi) \int_{-\pi}^{\pi} [2\pi\delta(w) + \pi\delta(w - \pi/2) + \pi\delta(w + \pi/2)] e^{jwn} dw \\ &= e^{j0} + (1/2)e^{j(\pi/2)n} + (1/2)e^{-j(\pi/2)n} \\ &= 1 + \cos(\pi n/2) \end{aligned}$$

(b) Using the transform synthesis equation (5.8)

$$\begin{aligned} x_2[n] &= (1/2\pi) \int_{-\pi}^{\pi} X_2(e^{jw}) e^{jwn} dw \\ &= -(1/2\pi) \int_{-\pi}^0 2je^{jwn} dw + (1/2\pi) \int_0^{\pi} 2je^{jwn} dw \\ &= (j/\pi) \left[-\frac{1 - e^{-j\pi n}}{jn} + \frac{e^{j\pi n} - 1}{jn} \right] \\ &= -(4/(n\pi)) \sin^2(n\pi/2) \end{aligned}$$

Problem 5.41

5.41 Let $\tilde{x}[n]$ be a periodic signal with period N . A finite-duration signal $x[n]$ is related to $\tilde{x}[n]$ through

$$x[n] = \begin{cases} \tilde{x}[n], & n_0 \leq n \leq n_0 + N - 1 \\ 0, & \text{otherwise} \end{cases},$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n}$$

for some integer n_0 . That is, $x[n]$ is equal to $\tilde{x}[n]$ over one period and zero elsewhere.

(a) If $\tilde{x}[n]$ has Fourier series coefficients a_k and $x[n]$ has Fourier transform $X(e^{j\omega})$, show that

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

regardless of the value of n_0 .

(b) Consider the following two signals:

$$x[n] = u[n] - u[n - 5]$$

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n - kN]$$

where N is a positive integer. Let a_k denote the Fourier coefficients of $\tilde{x}[n]$ and let $X(e^{j\omega})$ denote the Fourier transform of $x[n]$.

(i) Determine a closed-form expression for $X(e^{j\omega})$.

Problem 5.41



(b) Consider the following two signals:

$$x[n] = u[n] - u[n - 5]$$

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n - kN]$$

where N is a positive integer. Let a_k denote the Fourier coefficients of $\tilde{x}[n]$ and let $X(e^{j\omega})$ denote the Fourier transform of $x[n]$.

- (i) Determine a closed-form expression for $X(e^{j\omega})$.
- (ii) Using the result of part (i), determine an expression for the Fourier coefficients a_k .

Answer 5.41

5.41. (a) The Fourier transform $X(e^{j\omega})$ of the signal $x[n]$ is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=n_0}^{n_0+N-1} x[n]e^{-j\omega n}$$

Therefore,

$$X(e^{j2\pi k/N}) = \sum_{n=n_0}^{n_0+N-1} x[n]e^{-j(2\pi/N)kn}$$

Now, we may write the expression for the FS coefficients of $x[n]$ $\tilde{x}[n]$ as

$$a_k = \frac{1}{N} \sum_{\langle n \rangle} x[n]e^{-j(2\pi/N)kn} = \frac{1}{N} \sum_{n=n_0}^{n_0+N-1} x[n]e^{-j(2\pi/N)kn}$$

(Because $x[n] = \tilde{x}[n]$ in the range $n_0 \leq n \leq n_0 + N - 1$). Comparing the above equation with eq. (S5.41-1), we get

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$